

Goleta Beach Sediment Dumping Conservation Plan



ENV S 137

Winter 2026

By: Rebecca Sandoval, Zoe Ladd, Kate Bierwagen, and Quincy Wedbush
University of California, Santa Barbara, Environmental Studies Department
Professor Dawn A. Murray, PhD

Executive Summary

This conservation plan aims to explore a more comprehensive plan for sediment dumping off of Goleta Beach. Currently, Santa Barbara County has a beach nourishment and emergency sediment plan that allows them to dump large amounts of sediment off Goleta Beach. The sediment is dredged from other areas in the county and dumped in the ocean or added to widen the beach. Our goals are to ensure human access to a healthy beach and ocean and to safeguard the marine life that could be impacted by this dumping. In accomplishing these goals, we intend to bring justice to the community of Goleta as their access to healthy water to swim in and fish out of has been jeopardized.

In order to make this plan successful, a team of conservation planners will work closely with marine biologists, coastal engineers, sedimentologists, local communities, the Santa Barbara government, Indigenous groups, and non-governmental organizations. The main objectives are categorized as environmental, economic, or social. The environmental objectives are to maintain healthy populations of intertidal species and trophic connectivity, to mitigate sediment driven stressors on marine ecosystems, and to monitor and limit bioaccumulation of toxins in seafood. The economic objective is to minimize financial losses on a local business as it is located directly on Goleta Beach, right next to the dumping operations. The social objective is to safeguard water quality for human recreation and use.

The planned environmental targets are to maintain or improve intertidal population trends weeks to months after sediment dumping events and storms, maintain benthic invertebrate populations, set up mandated rounds of testing in the water, and report maintained or reduced toxin levels in sample catches months after dumping event. These targets are set to a duration of weeks- months and 10 years. The economic target is to maintain the same number of restaurant visitors over the course of sediment dumping. The social target is to report water quality consistent or with lower levels of contaminants of interest weeks and months after dumping.

For these targets to be achieved, we have devised specific strategies for each target, and through a multi-criteria decision analysis, the team is able to prioritize the most impactful, effective strategy. The highest ranking strategies include riparian improvement to enhance natural sediment flow, use of existing debris basins, more routine sediment disposal and distribution, and further research on sediment samples.

TABLE OF CONTENTS

Executive Summary.....	2
Project Scope.....	4
Project Vision.....	4
Planning Context.....	4
Introduction.....	4
Planning Team and Management Process.....	10
Decision Makers and Target Audience.....	10
Planning Team.....	10
Decisions, opportunities, sideboards, and constraints (optional).....	14
Fundamental Objectives and Desired Outcomes.....	14
Conservation Features.....	18
Situation Analysis.....	19
Range of Strategies.....	22
Strategy Selection.....	25
Multiple Criteria Decision Analysis.....	28
Data and Knowledge.....	31
Risks and Uncertainties.....	32
Risk 1: Lack of Data and Existing Science (Severe).....	32
Risk 2: Climate Change and Sea Level Rise (Severe).....	32
Conclusion.....	33
Bibliography.....	34

Project Scope

The spatial scope of this project is Santa Barbara County, CA with a primary focus on Goleta Beach and Goleta's watersheds. The project's objective is to maintain the biodiversity health of Goleta Beach and the surrounding waters, which Santa Barbara County encompasses. Particularly, the project aims to inform and facilitate the coastal ecosystem's recovery after large sediment dumping events, which would include short-term monitoring weeks to months after dumping and a long-term ecological study that would be up to 10 years.

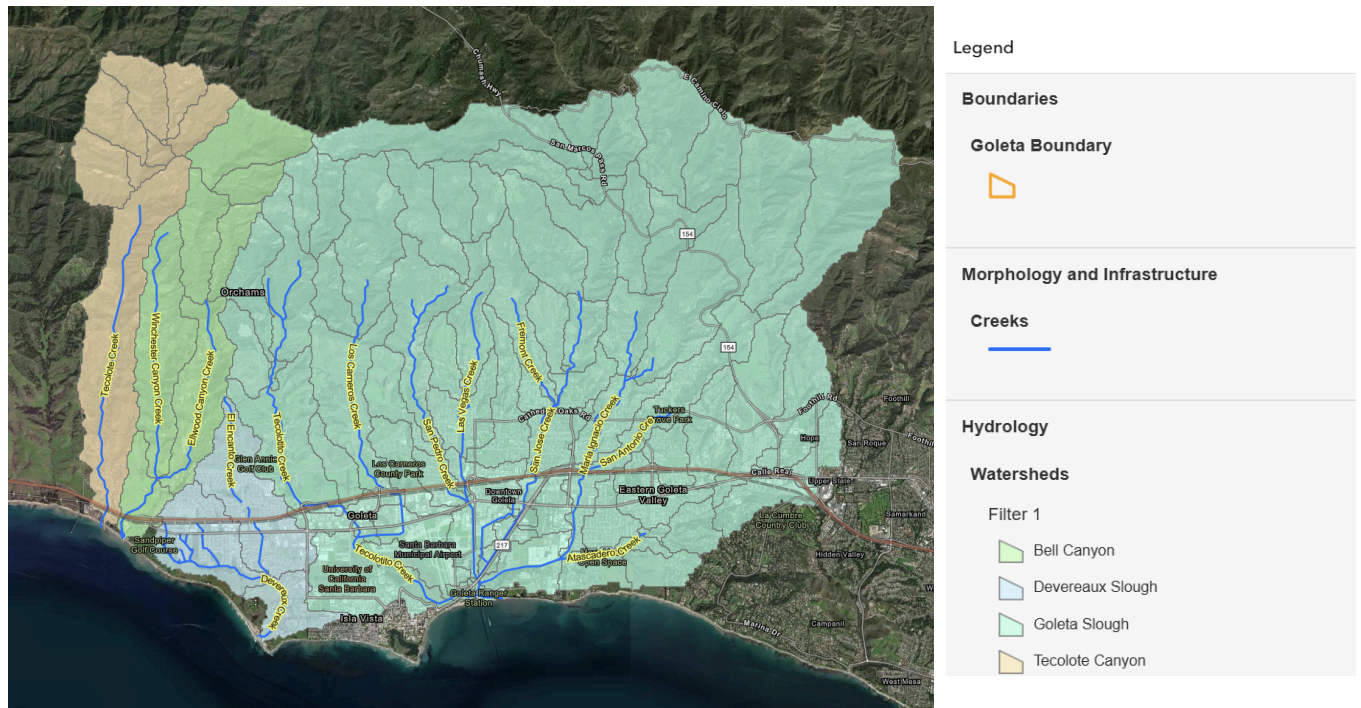
Project Vision

Our project vision is to have a healthy, resilient coastal marine ecosystem in Santa Barbara County where biodiversity thrives and people have access to a clean Goleta Beach.

Planning Context

Introduction

Our project will focus on the Goleta Beach sediment dumping, where the County of Santa Barbara has previously and is currently dumping sediments on Goleta Beach, causing unknown ecological effects on its greater marine coastal environment (Goleta Beach County Park to Partially Close for Emergency Beach Operations - Edhat, 2026). We are focusing on Goleta Beach, with the whole Santa Barbara county in mind as the sediment is moved throughout the county and impacts the whole community of Santa Barbara (Warrick et al., 2025). This sediment dumping calls for a more comprehensive management plan due to the possible adverse consequences on the coastal ecosystem. The current sediment deposition plans do not fully consider ecological repercussions and do not incorporate public input, creating an environmental injustice to the community of Goleta (*Santa Barbara County Age Characteristics*, n.d.).



Literature Review

Goleta Beach is one of the most visited parks in Santa Barbara County that is located adjacent to the University of California, Santa Barbara, getting roughly 1.5 million visitors a year (*Staff Report: Application No. 4-14-0909 (Goleta Beach Nourishment)*, 2015). The beach has many amenities including a playground, restaurant, picnic areas, and BBQ grills. Fishing, swimming, and boating are common activities at the park (*Goleta Beach Park*, n.d.).

The Santa Barbara Channel is a unique area of biodiversity and marine life, with a variety of intertidal and subtidal life. With giant kelp forests, rocky reefs, sandy seafloor and beaches, intertidal zones, and pelagic zones, there are innumerable interactions going on between the abiotic and biotic factors (Fernandez, 2022). There is a significant gap in existing scientific research to pinpoint the effects of sedimentation and dredging on marine life because of the intricate and numerous interactions between different species and water conditions.

The Goleta Slough connects water and sediment from the whole Goleta watershed and runs into Goleta Beach, so the beach has seen natural increases and decreases in sedimentation due to winter storm events and dry summers. Additionally, the beach has had significant erosion, prompting regular beach nourishment projects to maintain recreational access and safeguard infrastructure (*Goleta Slough Resilience Report*, 2023). Thus, humans have increased the sedimentation loads into the beach beyond natural levels through beach nourishment and emergency debris deposits from around the county after storm events (*Beach Operations*, 2026). Starting around 1994, the Goleta Slough Management Program corresponded with Santa Barbara County Flood Control and Water Conservation District began dredging the Slough and became routine in response to prevent flood risk. More recently, extreme weather events and wildfires have significantly increased sediment loads in Santa Barbara's watersheds, prompting emergency dredging operations. A major turning point occurred following the 2017 Thomas Fire and the catastrophic Montecito debris flows in 2018. These events generated massive quantities of sediment that accumulated in creeks and debris basins, requiring rapid removal to restore flood protection capacity. Under emergency permits, approximately 420,000 cubic yards of sediment were removed and transported to beaches including Goleta Beach and Carpinteria City Beach for

disposal; under these emergency operations, there are no limits on the quantity of large rocks or boulders deposited, even though these materials can significantly impact beach environments. More recently, beginning on February 2nd, 2026 sediment was hauled from Goleta Slough and distributed to Goleta Beach as an emergency storm response to remove excess sediment accumulated in flood-channels in Goleta (*Goleta Beach Section to Close for Operations | News List | Goleta, CA*, n.d.).

Dredging and Fish

Dredging is the removal of sediments and debris from the bottom of lakes, rivers, and other water bodies. It is intended to maintain important waterways and routinely happens in channels and harbors (National Oceanic and Atmospheric Administration, 2024). But, dredging can have a wide range of impacts on local aquatic and marine life. Wenger et al. (2017) performed a meta-analysis of different papers and existing studies related to dredging stressors on fish. They were able to categorize the stresses into six main impacts associated with dredging: habitat loss, hydraulic entrainment, release of contaminants, sedimentation, suspended sediment and underwater noise. The results of their linear mixed-effects model indicated that the effect of dredging was significantly influenced by the type of stress, the life-history stage during exposure, and the type of response the fish exhibited.

There were 4 response types exhibited by fish across the studies: behavioral changes, physical damage, physiological stress and sublethal responses, and mortality. Behavioral responses include avoidance of a stressor, reduced foraging rate or changes to habitat association. One of the most commonly observed behavioral changes by fish is the avoidance of turbid water. Under short-term circumstances of there being elevated suspended sediment, the avoidance behavior ceases once the disturbance is removed or if the fish becomes acclimated (Berg & Northcote, 1985). But, there have been instances where long-term elevated turbidity results in the disappearance of a species. Appleby & Scarratt 1989 found that dredging caused habitat degradation, increased sediments suspended, and contaminant release, which disrupted a key spawning ground for mackerel in the Sea of Marmara. Thus, the disappearance of mackerel in these waters was attributed to dredging (J.A. Appleby & D.J. Scarratt, 1989).

Physical damage is another response exhibited by fish after dredging and dumping events. This response primarily refers to the physical damage caused by the sediments, which refers to gill damage, skin abrasions or changes in development. For gill damage, the sediment can easily clog their gills and reduce their efficiency. This is especially important for smaller fish and larvae as the spaces between lamellae (part of the gill structure) are smaller in larvae. Wenger found that across studies any reduction in gill efficiency can have further stresses on the fish via difficulty with respiratory ability, nitrogenous excretion and ion exchange. In addition, another response by fish is physiological stress and sublethal responses, which they identified as changes in hormone levels, reduced growth rate, organ function or developmental abnormalities. And, the last response seen was increased mortality directly or indirectly through lower hatching success.

Again, the responses seen in fish depend on the species and its physiology, and the life stage of the fish. The meta analysis for one found that adult and juvenile fish observed physical damage and physiological impacts most frequently, respectively, while catadromous fishes were most closely associated with behavioural effects (Wenger et al., 2017). Across the studies they analyzed, the data also showed that larvae and eggs record lethal impacts more frequently than other life stages. These findings are important to consider for future research on the fish species near Goleta Beach to reveal how fish are affected by the consistent dumping. Another important finding was that the type of responses recorded for fish were similar across freshwater, estuarine, and marine species, meaning that dredging stressor studies on a range of species can be combined to develop general management guidelines for both marine and freshwater environments.

Kelp

As previously mentioned, the Santa Barbara Channel is home to biodiverse kelp forests. The Giant Kelp (*Macrocystis pyrifera*) in these nearshore ecosystems provides integral habitat for a wide range of aquatic organisms. Under natural conditions, sediment can be beneficial to primary producers like kelp, but when sediment loads are increased through human activity, negative impacts are more notable. Increased sedimentation poses particular risk to kelps at early life stages. Kelps release microscopic zoospore which can travel for up to three days before

settling and attaching to the substrate. Kelp zoospores are smaller than a grain of sand (less than 10 μm long). Deposited sediment can obstruct the settlement of zoospores by physically covering the substrate and reducing the zoospores' access to it. Deposited sediment can also smother the zoospores and obstruct respiration, photosynthesis, and the ability to uptake nutrients (Picard et al., 2022).

Picard et al. conducted a study which focused on *Alaria esculenta* and *Saccharina latissima* which are both found in colder waters than the Santa Barbara Channel. They placed spores of each species in containers with different levels of sediment (No sediment (control), low sediment, medium sediment, and high sediment). They measured three key factors to kelp recruitment: spore attachment, gametophyte density, and spore development success. They found that across both species, more sediment resulted in fewer gametophytes to germinate, indicating that high levels of sedimentation can interfere with early stages of growth in kelp. This study is on kelp species in the Gulf of St. Lawrence, Canada, neither of which are found in the Santa Barbara Channel. These findings, however, highlight the need for more research on the effects of sediment disposal on Giant Kelp (*Macrocystis pyrifera*).

Bat Star

Dong et al. conducted a study in 2025 to determine the molecular response of the *Patiria pectinifera* bat star to sediment burial. They analyzed the gene expression variations of these echinoderms to determine their stress responses. During sedimentation events, benthic organisms suffer from hypoxia stress (a lack of oxygen). They found that in response to increasing levels of physiological stress for *Patiria pectinifera* during sediment influx, metabolic changes occur and their immune system defenses are weakened. The focal species of this study, the *Patiria pectinifera*, is a type of bat star found in the northwest Pacific ocean and therefore is not found in the Santa Barbara Channel. The *Patiria miniata*, however, is a closely related species which is common in the Santa Barbara Channel. More research is needed to determine if the *Patiria miniata* have a similar reaction to sediment deposition as the *Patiria pectinifera*.

In summary, the Santa Barbara Channel has a wealth of biodiversity, and the coastal ecosystem off of Goleta Beach is part of that community. Lengthy analyses and studies have

been conducted to look at specific impacts of sedimentation events on species, but there is no full picture of the intricacies within the marine ecosystem. For instance, in investigating the impacts on fish, the meta analysis looked at over a few hundred different studies, and fish are only a small part of the life found off of the coast of Goleta Beach. The Channel has seagrasses, macroalgae, molluscs, crustaceans, echinoderms, bryozoans, not to mention marine mammals and a wide array of microbial life. Thus, to safeguard these species and fully understand how dredging impacts them and their interactions, there is more research needed.

Planning Team and Management Process

Decision Makers and Target Audience

The decision makers for the comprehensive management plan of the Goleta Beach sediment dumping mainly include officials from the County of Santa Barbara, specifically those in the Flood Control Division. The County is responsible for drafting up the official plan. Local community members, non-governmental organizations, and consultant scientists would to some extent contribute to the decision making by offering specialized information. Larger State and Federal agencies like the California Coastal Commission, Regional Water Quality Control Board and California Department of Fish and Wildlife, and U.S. Army Corps of Engineers would also be integral in approving the necessary permits to alter the sediment dumping operations.

Planning Team

A planning team coordinates and implements a conservation plan. This team, whose members are listed below in **Table 1**, has the expertise and resources to craft a comprehensive conservation plan. They create an inclusive list of fundamental objectives and are responsible for implementing and monitoring the steps needed to address each objective. **Table 1** identifies their strengths, entities, and role in making the conservation plan. The planners include representatives from government agencies, non-governmental organizations, community members, and scientists.

Member	Skillset	Organization	Role
---------------	-----------------	---------------------	-------------

Specialist or Planner of Flood Control District	Knowledge of ongoing flood control plans Familiarity with procedures	County of Santa Barbara - Flood Control District	Inform alternative plans Justify dumping site locations
NGOs	Funding Management expertise Knowledge of permitting required	Santa Barbara Channelkeeper Environmental Defense Center	Spread awareness Support implementation plans with funding, legal, and outreach capabilities
Marine Specialist	Marine biology expertise Intertidal and subtidal ecology knowledge	UCSB Researcher, US Fish and Wildlife Scientist or Scientific Consultant from Private Company, UCSB LTER	Quantify effect(s) on marine life off of coast Outline any species that are vulnerable
Sedimentologist or Coastal Engineer	Coastal erosion and beach sedimentation knowledge Knowledge of important sediment properties	UCSB Researcher, US Fish and Wildlife Scientist or Scientific Consultant from Private Company	Quantify how beach erosion and sedimentation is impacted

			Assess alternative dump sites and dredging effects
Community Members	Bring cultural and local perspective and experiences Understand local values and concerns	Goleta and Santa Barbara Residents	Inform/educate other community members on dumping effects and water quality of the beach

Table 1: Planning team members, their skillsets, and roles

Shareholders/Community Members

A shareholder is a group or individual who is invested in the outcomes of the project or impacted by the scope of it and whose participation is needed for thoroughness and effectiveness of the project. The community more widely is any party that values the conservation and health of Goleta Beach and its coastal ecosystem. They would still be impacted by the scope of the project but the extent to which they interact with the Goleta Beach varies. Different constituents may find value in different aspects of Goleta Beach such as cultural, economic, recreational, or subsistence. The community/interested partners/shareholders involved in our project, listed in **Table 2**, include non-governmental organizations, county and municipal management agencies, Indigenous groups, scientific agencies, private business owners, and local community members who visit the beach.

Member	Organization	Role
Santa Barbara Channelkeeper	Environmental Non-Profit	Monitor water quality impacts; advocate for stronger environmental protections

Santa Barbara County Flood Control	Country Government Agency	Removes sediment from channels; responsible for sediment transport decisions
Goleta Community Members/ UCSB Students	Local Community	Affected stakeholders; public voice; can participate in monitoring and outreach
California Coastal Commission	State Regulatory Agency	Approves and oversees coastal development and sediment placement permits
Recreational Fishermen	Public Stakeholder Group	Directly impacted by water quality and habitat changes
Recreational Beach-Goers	Public Stakeholder Group	Affected by water clarity, safety, and beach conditions
UCSB Cheadle Center	University Research Facility	Provides scientific data on slough ecology and habitat impacts
IV and SB Surfrider	Environmental Non-profit	Water quality monitoring; mobilizing public awareness
County of Santa Barbara	Local Government	Oversees project logistics and management
Businesses Groups in the Goleta Slough	Private Sector	Economically affected by ecosystem health
Barbareño Band of Chumash Indians	Tribal Government	Cultural stakeholder; protect ancestral lands and marine resources

Beach SMART	Education Program	Educates public about coastal systems and beach management
-------------	-------------------	--

Table 2 - List of shareholders, their organizations, and roles

Decisions, opportunities, sideboards, and constraints (optional)

There are multiple laws, agencies, and management organizations in place that have oversight of activities that influence the Goleta Beach Conservation Plan. The California Department of Fish and Wildlife and California Coastal Commission oversee a 384 acre space of the Goleta Slough that is designated as a marine protected area, in which “it unlawful to injure, damage, take, or possess any living, geological, or cultural marine resource” (*California Code of Regulations*, Title 13, § 632b; California Department of Fish and Wildlife, n.d.). These protections could limit activities within Goleta Slough and at Goleta Beach that could disturb marine habitats or alter natural coastal processes. Santa Barbara Channelkeeper is already working to mitigate sediment disposal stressors and operations at Goleta Beach. Working together with these organizations can provide additional resources for this conservation plan.

Within the City of Goleta and Santa Barbara County, grading and sediment-handling must comply with local Grading, Erosion and Sediment Control ordinances, NPDES stormwater requirements, and regional sediment-management policies administered by BEACON. The Goleta Slough management plans and the City/County Local Coastal Programs designate the Slough and adjacent habitats as protected, constraining where sediment can be placed (BEACON; State Water Resources Control Board).

Additionally, numerous pipelines from SoCalGas and the Goleta Sanitary District fall within the spatial boundaries of the conservation area, which may influence sediment dynamics and impose infrastructure constraints on restoration and management actions.

Fundamental Objectives and Desired Outcomes

The project’s fundamental objectives given below (**Table 3**) represent the conservation plan’s distinct strategies by outlining the focuses and goals we plan to achieve based on different shareholder interests. The objectives are broken down into specific categories including

environmental, social, and economic. Within the environmental category, there are three subcategories: biodiversity, ecological processes, and ecosystem services.

Goleta Beach	Fundamental Objectives	Features	Indicators	Targets
	<i>What & direction</i>	<i>What can be measured</i>	<i>How will 'what' be measured</i>	<i>Goal and time</i>
Environmental - Biodiversity	Maintain healthy populations of intertidal species	Intertidal species populations	Biodiversity level indexes, number of initial intertidal species within populations	Maintain or improve intertidal population trends in the weeks to months following sediment dumping events and storms (short-term), and sustain stable or increasing population trends over a 10-year period (long-term).
Environmental - Ecological Processes	Maintain trophic connectivity in intertidal zone	Benthic invertebrate populations	Use of grab samplers by oceanographers (US EPA, 2020)	Maintain benthic invertebrate populations weeks to months after sediment dumping events and storms (short-term) and sustain or increase benthic invertebrate population over 10

				years (long-term) (Tauran et al., 2025).
	Mitigate sediment driven stressors on the intertidal zone ecosystems	Contaminants and Suspended sediment	Water testing of contaminants and suspended sediments in water	Set up mandated round of testing (turbidity sensor-in situ nutrient sensor-multiparameter probe) before and after dumping
Environmental - Ecosystem Services	Monitor and limit bioaccumulation of toxins in seafood	Metal and toxin percentages in catch	Toxin levels in sample catches after sediment disposal	Report maintained or reduced toxin levels in sample catches months after dumping event.
Social: Human - Health and Equity	Safeguard water quality for human recreation and use	Water quality	Concentration of Fecal Indicator Bacteria (FIB) and Volatile Organic Compounds (VOCs)	Report water quality consistent or with lower levels of contaminants of interest weeks and months after dumping events (Li et al., 2020).
Economic	Minimize financial losses on local business	Financial Losses at Restaurants	Number of Restaurant Visitors	Maintain the same number of restaurant visitors over the course of sediment

				dumping.
--	--	--	--	----------

Table 3- Fundamental Objectives

Biodiversity

The central biodiversity target in this plan is to maintain healthy populations of intertidal species that live off of Goleta Beach. A primary concern of this plan is the depletion and harm to the marine species that live right off of the coast, so maintaining the high level of biodiversity in this ecosystem is a critical objective of minimizing the risk of sediment dumping. The Santa Barbara Channel is a unique hub of biodiversity, and sediment dumping off of one beach could have cascading effects on the species' lives downstream of the beach.

Ecological Processes

Of the fundamental objectives that are environmental, there is one fundamental objective targeting 'ecological processes,' and it aims to maintain trophic connectivity in the intertidal zone. Generally speaking, trophic connectivity is crucial in intertidal and pelagic zones. We are concerned about the cascading effects of sediment dumping, so to ensure healthy recovery of the ecosystems we would hope to maintain the trophic connectivity.

Ecosystem Services

Ecosystem services are services that the beach provides to improve the quality of life for humans. Here the main ecosystem service that the beach provides is food, as the beach offers significant amounts of seafood to local fishermen. The fundamental objective is thus to monitor and limit bioaccumulation of toxins in seafood. This would safeguard the community's access to food at the beach.

Social

Social fundamental objectives aim to address the human and social needs of the community, in this case those who interact most with Goleta Beach. The two fundamental objectives aim to safeguard the food source that Goleta Beach provides, i.e. fish, and the recreational activities common at Goleta Beach. While educational activities could be

implemented, we aim to achieve a scientific database on what the water quality looks like before and after the dumping. To maintain healthy recreation and fishing, testing the water frequently for contaminants and toxins will directly inform the safety of interacting with the beach.

Economic

Economic objectives are created to address different shareholders and boost the local economy. There is a new restaurant right on Goleta Beach, so the fundamental objective here is to minimize the financial losses. The concern is that the closures associated with the beach, that arise from sediment dumping, would affect the restaurant's finances.

Conservation Features

The fundamental objectives are each associated with their own features, indicators, and targets that demonstrate the steps needed to achieve each fundamental objective as shown in **Table 3**. Features are the measurable aspect of the objective. Indicators refer to the action/method in which the feature will be measured. Targets are the goal and timeframe in which the goal will be achieved. These factors are needed to make clear that our fundamental objectives contain a feasible, actionable goal.

For the fundamental objective of 'maintain healthy populations of intertidal species,' the feature that will be measured is intertidal species populations and the indicator is to use biodiversity level indexes to approximate the number of initial intertidal species within populations. Our goal is to maintain or improve intertidal population trends weeks to months after sediment dumping events and storms. For the fundamental objective of 'maintain trophic connectivity in intertidal zone,' the feature that will be measured is benthic invertebrate population and the indicator is to use grab samplers. Our goal is to maintain benthic invertebrate populations from weeks to months after sediment dumping events and storms. For the second fundamental objective targeting an ecological process, 'mitigating sediment driven stressors on the intertidal zone ecosystems,' the feature that will be measured is contaminants and suspended sediment and the indicator is water testing of contaminants and suspended sediments in water.

Our goal is to set up mandated rounds of testing (turbidity sensor-in situ nutrient sensor-multiparameter probe) before and after dumping. For the fundamental objective of 'monitor and limit bioaccumulation of toxins in seafood,' the feature that will be measured is metal and toxin percentages in catch and the indicator is toxin levels in sample catches after sediment disposal. Our goal is to report maintained or reduced toxin levels in sample catches months after the dumping event.

For the social fundamental objective of 'safeguard water quality for human recreation and use,' the feature that will be measured is water quality and the indicator is the concentration of Fecal Indicator Bacteria (FIB) and Volatile Organic Compounds (VOCs). Our goal is to report water quality consistent or with lower levels of contaminants of interest weeks and months after the dumping event.

For the economic fundamental objective of 'minimize financial losses on local business,' the feature that will be measured is financial losses at restaurants and the indicator is the number of restaurant visitors. Our goal is to maintain the same number of restaurant visitors over the course of sediment dumping.

Situation Analysis

A situation analysis is a tool made through Miradi Software that outlines the interactions and pieces of a conservation plan. This tool helps illustrate the web of interactions at play, and this visualization helps us better anticipate the factors and effects of our plan. As demonstrated below in **Figure 3**, we use a situation analysis for our plan to display our project scope, targets, the different threats, and the strategies we would use to achieve our goals. The scope is Santa Barbara County and Goleta Watershed, and our main targets within this scope are safeguarding Goleta Beach and maintaining marine health and life. Surrounding the scope are the different connections we made between threats (indirect and direct), biophysical factors, and potential strategies to protect the scope and targets within.

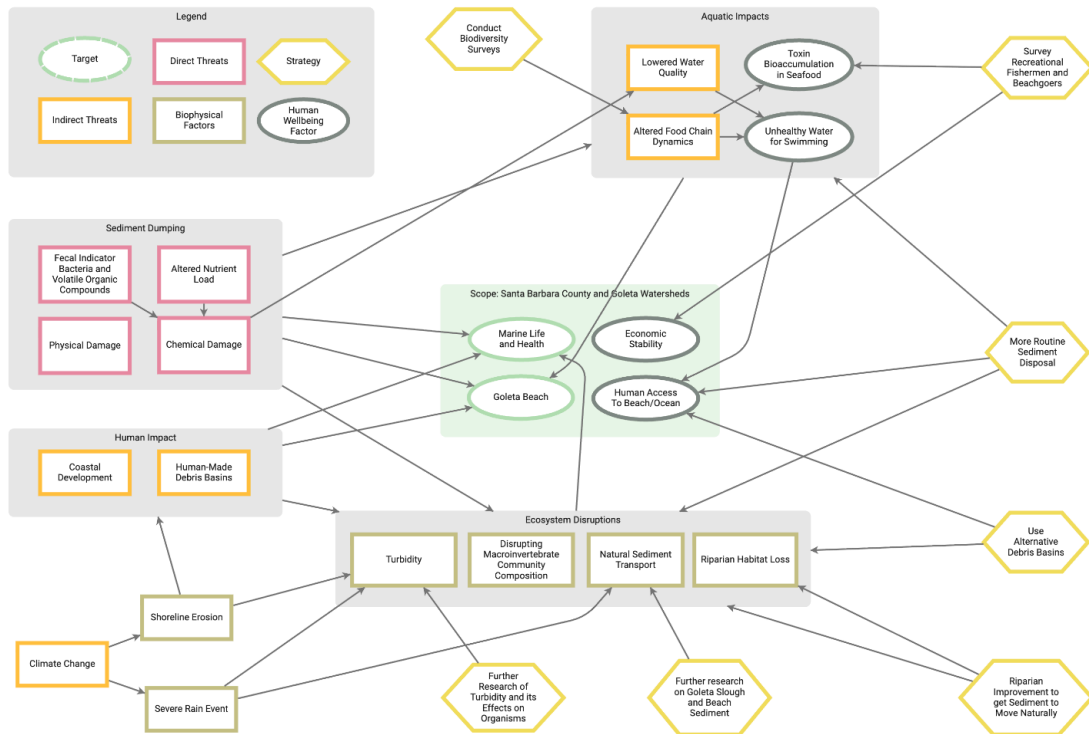


Figure 3- Situation Analysis displaying targets and human wellbeing factors, their direct threats, strategies, indirect threats, and biophysical factors

The main target of the conservation project is Goleta Beach and marine life and health, within the scope of Santa Barbara County and Goleta's watershed. Goleta Beach is currently the dump site for the county's emergency sediment management plan, which has put the water quality of the beach at risk and could threaten marine life off of the coast. In addition to the conservation target shown in green on the situation analysis, the human well-being targets are enclosed in dark green ovals within the project scope. The human well-being targets for this conservation project include the economic well-being of and access to the beach for the people in the region. Goleta Beach is one of the most visited parks in the county and has been a source of food and recreation for decades. Ensuring people have access to the beach and that local businesses remain economically stable are critical and at the center of our conservation plan. After establishing the scope of the project and the targets within, we were able to map out the threats to better understand which strategies would be most effective for our plan.

The threats to a healthy Goleta Beach and the ecosystem off its coast include the threats directly associated with sediment dumping: physical and chemical damage, increased fecal indicator bacteria and volatile organic compounds, and altered nutrient load. The threats are outlined in the situation analysis in red boxes. These threats connect to our main targets and to a series of other ripple effects. These direct threats connect to ‘Aquatic Impacts,’ including lowered water quality and altered food chain dynamics, which then jeopardize two human wellbeing factors: toxin bioaccumulation in seafood and unhealthy water for swimming and recreation. Additionally, the main threats connect to a series of ecosystem disruptions: changing turbidity, macroinvertebrate community composition, and natural sediment transport.

In addition to direct threats, there are indirect threats outlined in an orange rectangle on our situation analysis. These threats influence biophysical factors, human wellbeing factors, or contribute to other threats. For instance, there are two indirect threats grouped together under a box titled, ‘Human Impact.’ These are human activities that are impacting our targets and contributing to ecosystem disruptions. The two indirect threats here are coastal development and human-made debris basins. Under the ‘Aquatic Impacts’ box, as mentioned above, there are two indirect threats of lowered water quality and altered food chain dynamics that arise from our direct threats tied back to sediment dumping. And, our last indirect threat is climate change as it influences a myriad of other factors in our analysis. We connected it directly to influencing shoreline erosion and severe rain events, but it could likely be connected to each factor in our analysis.

The biophysical factors we identified are outlined in olive-colored rectangular boxes. These are the factors observed in the ecosystem that tie together the threats and the targets. In our case, there are many biophysical factors we anticipate as the ecosystem responds to massive sediment dumping. The ones we have identified include shoreline erosion, severe rain events, turbidity, macroinvertebrate community composition, natural sediment transport, and riparian habitat loss. Though, there are likely others at play that we have not identified.

Our chosen strategies are shown in yellow hexagons. The strategies are the possible solutions to mitigate the effects of the threats that are jeopardizing the health of Goleta Beach

and its coastal ecosystem. The threats we are addressing are all linked to sediment dumping since that is our focus. The strategies include conducting biodiversity surveys, further research on turbidity effects on organisms, survey recreational fishermen and beachgoers, more routine sediment disposal, use alternative debris basins, and riparian improvement to enhance natural sediment movement. More detailed explanations of the strategies can be found under **Table 4**.

Range of Strategies

Direct threats to Goleta Beach from sediment disposal include altered nutrient load, physical damage, chemical damage, and Fecal Indicator Bacteria (FIB) and Volatile Organic Compounds (VOCs). From these direct threats, a situation analysis was created highlighting human well-being factors, indirect threats, and biophysical factors related to these threats. Strategies this conservation plan proposes addressing these threats and subsequent impacts are summarized in **Table 4**, with more details on each strategy provided in the section located below.

Strategy	Description
Further research on Goleta Slough and Beach sediment	Invest in research of the particle composition, size, chemical, and biological composition of both sediments from the Goleta Slough and Goleta Beach.
Further research of turbidity effects on organisms	Invest in research of long-term turbidity effects on species composition and viability for organisms that inhabit Goleta Beach.
Survey recreational fishermen and beachgoers	Establish annual surveys of fishermen and beachgoers on observed effects of sediment dumping.
Use alternative debris basins	Use alternative debris basins to store sediment

	build up in flood channels.
Conduct biodiversity surveys	Conduct annual biodiversity surveys of intertidal organisms at Goleta Beach and before and after disposal sites.
Riparian improvement to enhance natural sediment flow	Engage in restoration projects to develop the riparian zones in key streams in the watershed. Enhanced zones will buffer sediment transport in storms.
More routine sediment disposal	Require more routine sediment operations rather than emergency operations to decrease magnitude and duration of sediment disposal at Goleta Beach.

Table 4 - Proposed strategies and their descriptions

No action

No intervention or action will likely lead to continued sediment disposal at Goleta Beach with unknown effects. Since these effects aren't well understood, there could be adverse negative impacts to the marine environment and contribute to reduced water quality at Goleta Beach.

Further research on sediment samples

Most beach nourishment activities in the past have proven that specific research to match the particle size and composition is needed to reduce adverse effects to the ecosystem (Pranzini et al., 2018; Tauran et al., 2025). Investing in research of grain size, shape, mineral composition, nutrients and metals, and foreign contaminants for both Goleta Beach and Goleta Slough can ensure that the natural beach sediment composition is similar to that of the Goleta Slough (Hannides et al., 2019). Additionally, ensuring sediment compatibility will help with maintaining trophic connectivity and biodiversity and protecting Goleta Beach's water quality.

Conduct further research on turbidity effects on organisms

The ecological impacts of turbidity generated by sediment disposal are not fully understood, especially for long-term exposure. Increased turbidity from sediment disposal can reduce light penetration, interfere with feeding in filter-feeding organisms, and alter benthic habitats (*Factsheet on Water Quality Parameters*, 2021). Research of turbidity effects on local species including mollusks, crustaceans, and filter-feeders, and other benthic invertebrates can help determine turbidity thresholds that organisms at Goleta Beach can tolerate. This research will help to maintain trophic connectivity and healthy intertidal populations.

Use alternative debris basins

Sediment buildup in flood channels around Goleta is a direct threat to the sediment disposal at Goleta Beach as it is the main driver of sediment disposal operations at Goleta Beach. Because sediment removal from these channels is necessary for flood management, the dredged material is often transported and dumped at Goleta Beach. Using alternative debris basins around Santa Barbara County would eliminate the sediment disposal at Goleta Beach, while maintaining required sediment removal. Eliminating the sediment disposal at Goleta Beach would protect the water quality, intertidal populations, and trophic connectivity.

Survey recreational fishermen and beachgoers

Lack of knowledge and long-term monitoring data is one of the main threats of sediment disposal at Goleta Beach. Establishing annual surveys of recreational fishermen and beachgoers can provide valuable observational data on environmental changes that may be overlooked by formal scientific monitoring. Due to the frequent use of Goleta Beach and the Santa Barbara Channel, these groups may notice changes in fish abundance, beach conditions, and wildlife abundance following sediment disposal events. Collecting this information can further advance management discussions of disposal events and subsequent ecological and recreational effects and improve adaptive management practices.

Conduct biodiversity surveys

Long-term effects of beach nourishment on intertidal invertebrates aren't well understood. Conducting annual biodiversity surveys in addition to surveys before and after

sediment disposal can identify changes in species abundance, diversity, and community composition over time. Gathering information on these factors can help adaptive management strategies for beach nourishment and sediment disposal at Goleta Beach. Additionally, establishing these surveys may provide additional data on climate change effects to coastal species.

Riparian improvement to enhance natural sediment flow

Restoration projects that improve the riparian zones around the streams in Goleta watershed can strengthen the natural sediment flow after. With more riparian vegetation, the banks can serve as a buffer zone, and filter sediments after storm events. Riparian forest buffer zones have been observed to filter nutrients, stabilize eroding banks, filter sediment from runoff, provide shade, shelter and food for aquatic organisms, and create recreational spaces, among other things (MacFarland et al., n.d.). Riparian buffers in the Goleta watershed could have similar benefits.

More routine sediment disposal

Goleta Beach is directly threatened by the altered nutrient load and fecal indicator bacteria as a result of sediment dumping from the magnitude that it is. Requiring more routine sediment disposal to reduce the magnitude and timing of the sediment disposal could greatly reduce these direct threats as having more routine sediment disposal mimics more natural sediment fluxes. Having more routine sediment disposal will also reduce sediment build up in flood channels and help maintain the infrastructure the County dredges sediment from (e.g., the Santa Barbara Airport). Lastly, more routine sediment disposal can reduce the physical beach alteration at Goleta Beach (plumes) arising from the large quantity of sediment being disposed of at times (Warrick et al., 2025).

Strategy Selection

To ensure a comprehensive comparison of strategies in this conservation plan, a Consequence Table (**Table 5**) and Multi-Criteria Decision Analysis (**Table 6**) were developed. These tables provide a structured framework for comparing advantages and disadvantages of

each strategy. They serve as visuals highlighting what strategies should be prioritized for implementation based on impacts to the conservation targets and fundamental objectives. The planning team will evaluate all strategies on how they will impact human well-being and ecological biodiversity at Goleta Beach.

Consequence Table

The Consequence table (**Table 5**) listed below summarizes the impact and tradeoffs that each proposed strategy has on the fundamental objectives (**Table 3**) created by the planning team. The table allows for a comparison of each strategy's impact on the fundamental objectives. The plus sign (+) indicates a positive impact on the fundamental objective. The slash (/) represents a neutral impact on the fundamental objective, neither negative nor positive impact. The negative sign (-) represents a negative impact on the fundamental objective. The positive, neutral, and negative impacts are summarized in the "Net Impact" column with a cumulative score and will be used to evaluate each strategy by the planning team.

Fundamental Objectives	Maintain healthy populations of intertidal species	Maintain trophic connectivity in intertidal zone	Mitigate sediment driven stressors on the intertidal zone ecosystems	Monitor and limit bioaccumulation of toxins in seafood source	Safeguard water quality for human recreation and use	Minimize financial losses on local business and the park	<u>Net Impact</u>
Indicators	Biodiversity level indexes, number of intertidal species within populations	Level of turbidity produced by sediment and population sizes	Levels of contaminants and suspended sediments in water	Toxin levels in sample catches	Concentration of Fecal Indicator Bacteria (FIB) and Volatile Organic Compounds (VOCs)	Number of Restaurant Visitors	

Strategies							
No Action	-	-	-	-	-	/	-5
Further Research on Sediment samples	(+)	(+)	(+)	(+)	(+)	/	+5
Further research of turbidity effects on organisms	(+)	(+)	(+)	(+)	/	/	+4
Use other debris basins	(+)	(+)	(+)	(+)	(+)	(+)	+6
Survey recreational fishermen and beachgoers	(+)	/	/	/	/	/	+2
Conduct biodiversity surveys	(+)	(+)	(+)	(+)	(+)	/	+5
Riparian improvement to enhance natural sediment flow	(+)	(+)	(+)	/	/	(+)	+4
More routine sediment disposal	(+)	(+)	/	/	-	/	+1

Table 5 - Consequence Table

The consequence table produced net impact scores of -5, +5, +4, +6, +2, +5, +4, and +1. Further discussion of these results is below:

1. Using other debris basins was the highest ranked strategy, receiving a net impact score of +6. Implementation of this strategy would eliminate all the direct effects of dumping sediment at Goleta Beach and would protect the marine environment. However, this strategy is not long term as the debris basins can only hold a set amount of sediment and also need to be cleared periodically.
2. The strategies to obtain a net impact score of +5, the second highest score, were further research on sediment samples and conducting biodiversity samples. Both of these strategies address the lack of existing science and data available, addressing the issues regarding lack of local, specific data on sediment dumping for the Santa Barbara County area. These two strategies address the environmental and social objectives involving marine life and human health.
3. Riparian improvement to enhance natural sediment flow and further research of turbidity effects on organisms both receive a net impact score of +4. Both of these strategies address the direct physical effects on marine organisms. Implementation of riparian improvement to enhance sediment flow through buffer zones would result in less dramatic sediment build up, easing the economic strain on sediment dumping. Further research on turbidity effects would also help address the lack of scientific knowledge of effects of sediment dumping. Both of these strategies address both environmental and economic objectives.
4. More routine sediment disposal received a net impact score of +1. This strategy solely focuses on impacts to marine organisms and would produce a negative economic impact.
5. No action received the lowest net impact score of -5. Implementation of this strategy would provide a negative impact to the majority of the fundamental objectives and would have detrimental effects on the community and environment at Goleta Beach.

Multiple Criteria Decision Analysis

Multiple Criteria Decision Analysis (MCDA) is a tool used for prioritizing actions and spatial conservation decisions. It also helps normalizing data, weighting criteria, & sensitivity analyses. There are two factors incorporated into the final scores: weight (1= least preference, 5=most preference) of each fundamental objective multiplied by rating (1= least fit, 10=best fit) of how each strategy option meets the fundamental objective. Each row in every strategy option is added together to create totals on the bottom, the higher numbers indicating more valuable strategies.

Fundamental objective	Strategies (1-10)	Further Research on Sediment samples	Further research of turbidity effects on organisms	Use existing debris basins	Survey recreational fishermen and beachgoers	Conduct biodiversity surveys	Riparian improvement to enhance natural sediment flow	More routine sediment disposal and distribution	No Action
	Weight (1-5)								
Maintain healthy populations of intertidal species	5	8(40)	9(45)	8(40)	4(20)	10(50)	7(35)	7(35)	1(5)
Maintain trophic connectivity in intertidal zone	4	8(32)	10(40)	8(32)	3(12)	10(40)	9(28)	7(28)	1(4)
Mitigate sediment driven stressors on the intertidal zone ecosystems	5	10(50)	10(50)	10(50)	3(15)	7(35)	10(50)	9(45)	1(5)
Monitor and limit bioaccumulation of toxins in seafood and protect food	3	9(27)	9(27)	6(18)	10(30)	5(15)	8(24)	7(21)	1(3)
Safeguard water	4	5(20)	1(4)	10(40)	8(32)	3(12)	8(32)	1(4)	1(4)

quality for human recreation and use									
Minimize financial losses on local business and the park	2	2(4)	3(6)	5(10)	10(20)	5(10)	5(10)	1(2)	1(2)
Total:		173	172	190	129	162	195	179	23

Table 6 - Multiple Criteria Decision Analysis for Goleta Beach Sediment Disposal

The MCDA prioritized management strategies by first weighing our objectives according to their significance to the biophysical outcomes and long-term ecosystem health. Objectives such as maintaining healthy populations of intertidal species and mitigating sediment-driven stressors were given the highest weights (5), reflecting our emphasis on the ecological concern of the sediment dumping. Additional objectives such as maintaining trophic connectivity and safeguarding water quality were also weighted highly, while economic considerations, such as minimizing financial losses to local businesses, were weighted lower. This weighting reflects the goal of the analysis, which was to evaluate strategies primarily through the lens of ecological science and long-term environmental stability rather than short-term economic convenience.

Based on these weighted objectives, several strategies emerged with the highest total scores. Riparian improvements to enhance natural sediment flow received the highest overall score (195), followed by the use of existing debris basins (190) and more routine sediment redistribution (179). These strategies were highly ranked because they directly address ecological drivers of the problem by reducing unnatural sediment concentration in sensitive intertidal habitats.

While completely relocating sediment away from the beach would likely prevent many of the ecological issues associated with dumping, such an option was outside the realistic management alternatives considered for the past 30 years. As a result, strategies focused on improving knowledge and ecosystem-based management ranked highly, as further research on sediment composition and turbidity effects scored 173 and 172. This is critical as a clearer scientific understanding of sediment impact can provide stronger and clearer evidence for why

current dumping practices are harmful. This information can ultimately support better management decisions and more effective long-term protection of the Goleta intertidal ecosystem.

Data and Knowledge

A wide range of data and knowledge is crucial to the success of this conservation plan, and because there are existing management plans for sediment dumping on Goleta Beach, we focused on investigating the gaps and flaws of these existing plans. Thus, the County of Santa Barbara is a significant resource for informing this plan. They have posted their plans, reported on beach closures, and have mandated water testing data that they have available online. There are associated reports with the sediment dumping plan, such as an Environmental Impact Report, but this report is available directly online as it was made more than a decade ago. Local newspapers have published papers reporting on beach closures and public opinion, which supplemented the governmental reports.

Moreover, this plan focuses on the possible ecological repercussions of the ongoing sediment dumping. So, much of the research has involved academic journals regarding relevant ecosystems and dredging. These sources were referred to us by a UCSB librarian and a policy associate from Santa Barbara Channelkeeper, Nate Irwin, who is familiar with the sediment dumping. Other academic literature was sourced through online searching on search engines like Google Scholars using key words such as ‘dredging.’ Additionally, a previous Bren project examined this topic of sediment dumping on Goleta Beach, so this was an informative resource to reference as they did research into the permitting of the plan and did in-person surveying of recreational beachgoers.

As much further research is needed, more data would be collected by biologists, ecologists, sedimentologists, geologists, and other specialists that would be able to generate more data on what is going on in the water exactly off of Goleta Beach, given the magnitude and frequency of sediment dumping.

Risks and Uncertainties

There are a number of risks and uncertainties associated with sediment dumping at Goleta Beach which our conservation plan aims to address. Some of these risks are outside the scope of what our conservation team is able to reasonably address. The increasingly severe impacts of climate change and the wide gaps in scientific knowledge of the ecological effects of sediment dumping are both severe risks and causes of uncertainty.

Risk 1: Lack of Data and Existing Science (Severe)

The lack of scientific data poses a significant threat to the ability to adequately protect the biodiversity of Goleta Beach. Decisions to dump large amounts of sediment at high volumes have been made at Goleta Beach for decades without a holistic understanding of what the immediate and long-term ecological effects may be. There is a pressing need for comprehensive documentation of present species and assessments of environmental changes before, during, and after sediment dumping events. The collection of data must be a priority from the start of these large-scale sediment dumping events to adequately monitor the ecological effects. As Nate Irwin (Policy Associate for Santa Barbara ChannelKeeper) wrote in a letter to BEACON in regards to their 2025 Draft Regional Coastal Adaptation Monitoring Program (RCAMP): “Without elevating ecological monitoring to the level of a critical priority it will be difficult to evaluate whether adaptation measures are sustaining the health of habitats, species, water quality, and ecosystem functions - key considerations in assessing changed conditions and evaluating project effectiveness.” (2025, September 12). Particularly, there is a lack of long-term research on the effects of sediment dumping and dredging, meaning that when these decisions are made there is no comprehensive understanding of the long-term effects on the diverse beach and aquatic ecosystems at Goleta Beach.

Risk 2: Climate Change and Sea Level Rise (Severe)

Climate change poses a significant threat to the long-term protection of Goleta Beach. Rising sea levels are expected to increase shoreline erosion rates, which may increase the need for beach nourishment to maintain the shoreline. Increased wave energy associated with higher sea levels may also accelerate the loss of nourished sediments, reducing the long-term

effectiveness of nourishment projects (*SLR Science & Consequences*, n.d.). In addition, more frequent and intense rainfall events may increase sediment accumulation in flood channels and raise the likelihood of debris flows. Increased wildfire risk associated with climate change is another concern, as post-fire conditions can increase sediment transport during storms and require additional sediment removal for debris management, as observed following the 2018 Montecito debris flow (Li et al., 2020).

Changes in precipitation patterns and coastal processes may also alter natural sediment transport and deposition dynamics, potentially increasing the demand for beach nourishment while reducing its long-term viability (*SLR Science & Consequences*, n.d.). Furthermore, more intense rainfall events can increase surface runoff carrying sediments, nutrients, and pollutants into coastal waters, which may degrade water quality and negatively impact marine ecosystems near Goleta Beach. Because the magnitude and timing of climate change impacts remain uncertain, these processes may increase sediment management demands, however it is difficult to predict how the Goleta Beach Conservation plan will be affected with these increased sediment management demands.

Conclusion

The current sediment management of Goleta Beach, while carefully thought through and reviewed, has large knowledge gaps that need to be addressed. There is no ongoing ecological study of how the sediment dumping affects the marine life off of Goleta Beach, and the repeated beach closures due to elevated fecal indicator bacteria after sediment dumping point to an insufficient, inequitable system. In this plan, the fundamental objectives to maintain healthy marine life and water quality offer the opportunity for the sediment management plan to reprioritize the ecosystem of Goleta Beach. And in safeguarding a healthy marine ecosystem, the people of Goleta will be rewarded with healthy water to swim in and fish from. These targets can be achieved with more water quality testing and ecological monitoring to guide alternative dumping strategies. The strategies mentioned in this plan offer promising alternatives, but with more research and a cost-benefit analysis, the County could fully investigate what is the most equitable, comprehensive, effective management plan.

Bibliography

- Barbareño Band of Chumash Indians*. (2021, January 27). Barbareño Band of Chumash Indians. <https://www.bbc-indians.com/>
- Beach Operations*. (2026, February 18). <https://www.countyofsb.org/3711/Beach-Operations>
- BeachSMART Experience*. (n.d.). Retrieved February 15, 2026, from <https://experience.arcgis.com/experience/778c07a081c54a15939f9ebd8cc8557a/page/Home>
- BEACON. (n.d.). *Who We Are*. The Beach Erosion Authority for Clean Oceans and Nourishment. Retrieved March 13, 2026, from <https://beacon.ca.gov/who-we-are/>
- Berg, L., & Northcote, T. G. (1985). Changes in Territorial, Gill-flaring, and Feeding Behavior in Juvenile Coho Salmon (*Oncorhynchus kisutch*) following Short-term Pulses of Suspended Sediment. *Canadian Journal of Fisheries and Aquatic Sciences*, 42(8), 1410–1417. <https://doi.org/10.1139/f85-176>
- Bren School of Environmental Science & Management. (2025). *DebrisEase: Final report*. University of California, Santa Barbara. https://bren.ucsb.edu/sites/default/files/2025-04/DebrisEase_FinalReport.pdf
- California Code of Regulations*, Title 14, § 632(b). (n.d.). <https://govt.westlaw.com/calregs/>
- California Department of Fish and Wildlife. (n.d.). *Goleta Slough State Marine Conservation Area (No-Take)*. Retrieved March 13, 2026, from <https://wildlife.ca.gov/Conservation/Marine/MPAs/Goleta-Slough>
- Comments Received on Sediment Deposition Projects at Goleta Beach and Carpinteria Beach*. (n.d.). Retrieved March 10, 2026, from <https://beacon.ca.gov/wp-content/uploads/2023/03/Comments-Received-on-Sediment-Deposition-Projects-at-Goleta-Beach-and-Carpinteria-Beach.pdf>
- Cox, R., O’Boyle, L., & Cytrynbaum, J. (2019). Imbricated Coastal Boulder Deposits are Formed by Storm Waves, and Can Preserve a Long-Term Storminess Record. *Scientific Reports*, 9(1), 10784. <https://doi.org/10.1038/s41598-019-47254-w>
- Dong, H., Wan, L., Wang, C., Sun, C., Wang, X., & Xu, L. (2025). Transcriptomic Responses of Blue Bat Star *Patiria pectinifera* to Sediment Burial. *International Journal of Molecular Sciences*, 26(11), 5208. <https://doi.org/10.3390/ijms26115208>

Effects of sediment on spore performance as a potential constraint on kelp distribution. (2022). *Marine Pollution Bulletin*, 185, 114336. <https://doi.org/10.1016/j.marpolbul.2022.114336>

Essink, K. (1999). Ecological effects of dumping of dredged sediments; options for management. *Journal of Coastal Conservation*, 5(1), 69–80. <https://doi.org/10.1007/BF02802741>

Factsheet on Water Quality Parameters. (2021, July). Environmental Protection Agency. https://www.epa.gov/system/files/documents/2021-07/parameter-factsheet_turbidity.pdf

Fernandez, S. (2022, November 15). *A Changing Sea Floor*. The Current. <https://news.ucsb.edu/2022/020774/changing-sea-floor>

Goleta Beach Park. (n.d.). Retrieved March 4, 2026, from <https://www.countyofsb.org/813/Goleta-Beach-Park>

Goleta Beach Section to Close for Operations | News List | Goleta, CA. (n.d.). Retrieved March 3, 2026, from <https://www.cityofgoleta.org/Home/Components/News/News/16433/3875>

Goleta Slough Resilience Report. (2023).

Goleta Surfing: A Point is Born. (2024, January 24). *Edhat*. <https://www.edhat.com/news/goleta-surfing-a-point-is-born/>

Hannides, A., Elko, N., & Humiston, K. (2019). The state of understanding of the effects of beach nourishment activities on coastal biogeochemical processes and conditions. *Shore and Beach*, 86, 46–57. <https://doi.org/10.34237/1008734>

J.A. Appleby & D.J. Scarratt. (1989). *Physical effects of suspended solids 0, on marine and estuarine fish and shellfish with special reference to ocean dumping: A Literature Review*. (Canadian Technical Report of . Fisheries and Aquatic Sciences No. 1681). Biological Sciences Branch Department of Fisheries and Oceans.

Li, D., Van De Werfhorst, L. C., Dunne, T., Devarajan, N., Ayala, T. G., & Holden, P. A. (2020). Surf zone microbiological water quality following emergency beach nourishment using sediments from a catastrophic debris flow. *Water Research*, 176, 115733. <https://doi.org/10.1016/j.watres.2020.115733>

MacFarland, K., Straight, R., & Dosskey, M. (n.d.). *Riparian Forest Buffers: An Agroforestry Practice*.

National Oceanic and Atmospheric Administration. (2024, June 16). *What is dredging?* <https://oceanservice.noaa.gov/facts/dredging.html>

- Oak, H. L. (2022). *Boulder beaches: A sedimentological study* [Thesis, Macquarie University].
<https://doi.org/10.25949/19438427.v1>
- Pranzini, E., Anfuso, G., & Muñoz-Perez, J. J. (2018). A probabilistic approach to borrow sediment selection in beach nourishment projects. *Coastal Engineering*, 139, 32–35.
<https://doi.org/10.1016/j.coastaleng.2018.05.001>
- Saengsupavanich, C., Pranzini, E., Ariffin, E. H., & Yun, L. S. (2023). Jeopardizing the environment with beach nourishment. *Science of The Total Environment*, 868, 161485.
<https://doi.org/10.1016/j.scitotenv.2023.161485>
- SANTA BARBARA COUNTY Age Characteristics. (n.d.). Retrieved March 11, 2026, from
https://www.sbcag.org/wp-content/uploads/2023/08/age_characteristics_report.pdf
- SLR Science & Consequences. (n.d.). Retrieved March 11, 2026, from
<https://www.coastal.ca.gov/climate/slr/science/>
- Staff report: Application No. 4-14-0909 (Goleta Beach nourishment) (2015).
<https://documents.coastal.ca.gov/reports/2015/7/w26d-7-2015.pdf>
- State Water Resources Control Board. (n.d.). *National Pollutant Discharge Elimination System (NPDES)—Wastewater* | California State Water Resources Control Board. Retrieved March 13, 2026, from https://www.waterboards.ca.gov/water_issues/programs/npdes/
- Tauran, A., Lavesque, N., Blanchet, H., Bujan, V., Gouillieux, B., Humbert, S., Lamarque, B., Latry, L., & de Montaudouin, X. (2025). Long-term impact of dredging and beach nourishment works on benthic communities. *Estuarine, Coastal and Shelf Science*, 313, 109119. <https://doi.org/10.1016/j.ecss.2024.109119>
- US EPA, O. (2020, July 20). *Benthic Indicator Research* [Overviews and Factsheets].
<https://www.epa.gov/water-research/benthic-indicator-research>
- Warrick, J. A., Stevens, A. W., & Tehranirad, B. (2025). Coastal fine-grained sediment plumes from beach nourishment near Santa Barbara, California. *Coastal Engineering Journal*, 67(3), 558–582. <https://doi.org/10.1080/21664250.2025.2497705>
- Wenger, A. S., Harvey, E., Wilson, S., Rawson, C., Newman, S. J., Clarke, D., Saunders, B. J., Browne, N., Travers, M. J., Mcilwain, J. L., Erftemeijer, P. L. A., Hobbs, J.-P. A., Mclean, D., Depczynski, M., & Evans, R. D. (2017). A critical analysis of the direct effects of dredging on fish. *Fish and Fisheries*, 18(5), 967–985. <https://doi.org/10.1111/faf.12218>

Wikipedia, A. at E. (2006). *English: Goleta Beach from UCSB cliffs, Santa Ynez Mountains in background*. [Graphic]. Transferred from en.wikipedia to Commons by Quadell using CommonsHelper. <https://commons.wikimedia.org/wiki/File:GoletaBeach.JPG>

Wooldridge, T., Henter, H. J., & Kohn, J. R. (2016). Effects of beach replenishment on intertidal invertebrates: A 15-month, eight beach study. *Estuarine, Coastal and Shelf Science*, 175, 24–33. <https://doi.org/10.1016/j.ecss.2016.03.018>